

Antoine Lopez

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PROFILE

Results-driven **Data Engineer** specialized in scalable distributed systems, high-throughput ETL pipelines, and cloud lakehouse architectures. Experienced in building robust streaming and batch processing workflows with Apache Spark, Kafka and Cloud.

EXPERIENCE

AI Scientist Intern

February 2026 – August 2026

ALTEN

Sèvres, France

- Developed an end-to-end framework for predictive system-failure modeling in complex Web3 financial networks (JEPA)
- Engineered an Ego-temporal network pipeline using PyTorch Geometric to capture multi-scale local node dynamics and neighborhood evolution (GCP BigQuery)
- Implemented a JEPA to predict future latent representations of network states (Python, PyTorch)
- Managed the end-to-end research pipeline, including formalizing the theoretical hypothesis, designing custom data ingestion for temporal graphs, and validating model robustness against historical system shocks.

Data Engineer Intern

May 2025 – August 2025

Société Générale

Val de Fontenay, France

- Designed and assessed software testing strategies in a complex environment
- Implemented new testing strategies tailored to requirements (Scala, Java, Spark, Apache Kafka)
- Developed a testing application for cloud data lake ingestion, including APIs, S3 buckets, and data processing pipelines (Python, REST API, AWS S3)

Software Developer Intern

January 2024 – February 2024

VINCI Energies — Cloud, Communication and Services

La Défense, France

- Developed a chatbot for client ticket management using Llama 2 (Python, Docker, Linux VM)

Junior Engineering Office Technician Intern

December 2022 – February 2023

VINCI Energies — AXIANS

Palaiseau, France

- Wrote procedures and processed Work Declarations and Notices of Intent to Commence Work
- Built an Excel database for managing customer contracts (Excel, VBA)

EDUCATION

ECE Paris

September 2021 – August 2026

Master's Degree in Engineering - Big Data & Artificial Intelligence | Cloud Option

Paris, France

Ajou University

August 2023 – January 2024

University Exchange Semester — Computer Science

Suwon, South Korea

French Baccalauréat | Maths, Physics-Chemistry, Computer Science

2021

PROJECTS

Wikipedia Real-Time Stream | PySpark, Databricks, SQL, Big Data

- Engineered a real-time data streaming and aggregation pipeline using PySpark and Databricks SQL.

Airport Data Extraction Pipeline | Scala, Functional Programming, ETL

- Implemented an automated ETL engine in Scala to parse, clean, and analyze high-volume airport metrics.

Hackathon AWS by IPPON | Cloud, AWS, Resilience

- Managed maintenance and scaling of a high-load platform on AWS under chaotic scenarios.

MusicRecognizer | MLOps, Docker, CI/CD, FastAudio

- Deployed an end-to-end audio recognition service with automated CI/CD and Docker containerization.

TECHNICAL SKILLS

Big Data & Streaming: Apache Spark (PySpark), Apache Kafka, Databricks, Hadoop, Data Lake

Cloud & Storage: AWS, GCP BigQuery, MySQL, MongoDB, Neo4J, Docker

Programming Languages: SQL, Scala, Python, Java

Data Engineering & Tools: ETL/ELT Pipelines, REST APIs, Git/GitHub, CI/CD, Agile (Scrum)

Certifications: AWS Academy Graduate, Google Cloud (BigQuery Insights)

LANGUAGES & INTERESTS

Languages: French (Native), English (Professional), Spanish (Intermediate)

Interests: Tech & AI Watch, Web3, Drawing, Running, Football, Hiking, Esport competitions